

LAYAN KAUSHIK

Data Analytics | Data Science | Machine Learning

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PERSONAL PROFILE

With a background in **Computer Science and Economics**, I am skilled at applying Machine Learning and Data Science tools, with a particular focus and experience as a **Research Assistant** working on *predictive analytics, causal inference, outlier detection, natural language processing, and data-driven decision-making*. My professional experience includes **2 years as a Data Analyst** and subsequently as a **Consultant at Mastercard**, where I concentrated on data-driven strategies and product development in the payment industry. I am currently seeking opportunities in the field of **Data Science and Analytics** to apply my skills in solving complex problems and driving innovation in the field.

EDUCATION

Duke University, Durham, NC, USA
M.S. Economics and Computation

Aug 2022 – May 2024 (Expected)
GPA: 3.89/4.0

Courses: Theory and Algorithm of Machine Learning (Decision Trees, Variable importance, SVM, Mixture Models and EM), Data Science (Regression, Pattern Mining, Outlier Detection), Elements of Deep Learning (MLP, CNN, RNNs, Reinforcement Learning), Advanced Algorithms, Advanced Econometrics (Statistical Theory, Causal Inference), Time Series Econometrics (ARIMA), Algorithmic Game Theory (Auction Design).

Indian Institute of Technology, Roorkee, India
B.Tech. Civil Engineering, First Division with Distinction

Jul 2016 – May 2020
GPA: 8.726/10

Courses: Computer Programming and Numerical Analysis using C++, Probability & Statistics, Data Mining, Linear Optimization.

WORK EXPERIENCE

Research Assistant, under Prof. Michael Pollmann, Duke University

May 2023- Present

- Integrating machine learning and statistical techniques to address measurement errors in causal models.
- Developed an R package to estimate treatment effects with measurement error in covariates and repeat measurements.
- The package focuses on settings where some of the covariates are measured with error and repeat measurements are available. Methods for high-dimensional covariates are also available.
- Improved data processing efficiency by integrating C++ with R, significantly cutting down runtime for large-scale analyses.
- Analyzed and replicated Matlab's Moore-Penrose Pseudoinverse function in R, comparing its efficiency with SVD-based calculations.
- **Technical Skills:** R, C++, Causal Inference, Machine Learning, Shell script (SLURM), LaTeX.

Research Assistant, under Prof. Tong Guo, The Fuqua School of Business

Jan 2024- Present

- Utilized Natural Language Processing and machine learning to analyze 1.29 million ADHD-related TikTok videos, to improve data integrity and classify advertised and organic content.
- Currently working on topic modeling to identify patterns in user engagement, focusing on uncovering prevalent themes and trends within ADHD-related discussions on TikTok.
- **Technical Skills:** Python, Machine Learning, Natural Language Processing, Web Scrapping, Topic Modeling.

Data Analyst & Consultant, Mastercard, India

Aug 2020- Jun 2022

- **Product Enhancement:** Enhanced Mastercard's Geographic Insights product for greater geospatial precision and integrated machine learning algorithms to meet client-specific requests, leading to increased client onboarding.
- **Predictive Modeling:** Analyzed and predicted the financial impacts of COVID-19 and natural disasters like hurricanes and earthquakes in different parts of the US using supervised machine learning algorithms on Mastercard transactional data.
- **Strategic Data Analysis:** Generated marketing campaign strategies for a major bank in Kuwait to increase their cross-border online sales using data-driven insights and test & learn strategies.

- **Business Consulting:** Provided partnership opportunities and innovation roadmaps tailored to clients' needs and interests under Mastercard's Startup Engagement Program, Start Path.
- **Automation:** Automated data delivery tasks for recurring clients, boosting client onboarding to Mastercard's Geographic Insights product by 50%.
- **Technical Skills:** Python, SQL, Big Data tools (Hadoop, PySpark, HIVE, Impala)
- **Soft Skills:** Critical Thinking, Teamwork, Presentation skills, Stakeholder Engagement, Ethical Judgment

ACADEMIC PROJECTS

Data Enhancement of Flood Insurance Claims dataset for Flood Damage Prediction, Independent Research, Duke University

- Presented a conference poster at the American Geophysical Union (AGU) 2023 conference.
- Improved Flood Insurance Claims dataset by methodically resolving data inconsistencies and systematically addressing missing values to ensure higher reliability for in-depth analysis.
- Enhanced mapping precision using Python geospatial libraries, generating detailed and more precise new geographic shapefiles for analysis.
- Enriched predictive models by incorporating demographic, geographic, and weather data, sharpening accuracy in flood damage estimations.
- Currently developing advanced outlier detection algorithms in Python to refine water-depth measurements within datasets, thereby enhancing the quality of data for predictive modeling.
- Currently developing a data pipeline to enable the creation of the new geographic unit for future updated data, designed to improve precision and support the use of advanced outlier detection algorithms.
- **Technical Skills:** Python, R, Geospatial Analysis, ETL, Data Science Workflow.

SKILLS

Programming Languages: Python, R, C++, Matlab, SQL, ML Frameworks: Pytorch

Miscellaneous: Stata, Shell (Bash/SLURM), LaTeX, Tableau, MS Excel, Git

Languages: English (fluent), Assamese (native speaker), Hindi (fluent)

HONORS AND AWARDS

Dean's Research Award for Master's Students, Duke University	2023
U.S. Army Corps of Engineers Institute for Water Resources Research Grant, USACE	2023
Duke Economics Master's Scholar Award, Duke University	2022
Annual Excellence Award, IIT Roorkee Heritage Foundation	2018 & 2019
IASc-INSANA Summer Research Fellowship, Indian Academy of Sciences	2018
Top 3 in State Mathematics Olympiad, Homi Bhabha Centre for Science Education	2014

VOLUNTEER WORK AND OTHER ACTIVITIES

Graduate Teaching Assistant, Duke University	<i>Jan 2023- Jul 2023</i>
Worked as a Graduate Teaching Assistant for the following courses: CS216: Everything Data (Spring 2023) and CS230: Discrete Math for CS (Summer 2023). Responsibilities included holding office hours, managing undergraduate TAs, creating assignments, delivering recitation lectures, and automating the grading process using Otter-Grader.	
Assistant Badminton Coach, Duke University	<i>Aug 2022- May 2023</i>
Provide regular training and coaching advice to players affiliated with Duke Badminton Club.	
Student Mentor, Student Mentorship Program, IIT Roorkee	<i>Jul 2018- May 2019</i>
Mentored a batch of 7 freshmen to help with their transition from high school to university.	
Undergraduate Teaching Assistant, IIT Roorkee	<i>Jul 2017- May 2018</i>
Worked as an Undergraduate Teaching Assistant for the following courses: Computer Programming in C++ (Spring 2018) and Solid Mechanics (Fall 2017). Responsibilities included holding office hours and lab sessions.	